MUSP-BENCH: ADVANCED MULTIMODAL BENCHMARKING OF MUSIC UNDERSTANDING ACROSS SCORE AND PERFORMANCE

Milan Liessens Dujardin[1,2] **Song-Ze Yu**[2] **Kevin Miao**[1]
[1] Bryel Labs [2] UC Berkeley
milan.ld@berkeley.edu vaclis@berkeley.edu kevin@bryel.ai

ABSTRACT

Musicians commonly communicate music through scores and performances. Scores encode musical intent, while performances realize it in sound. To investigate whether models can meaningfully engage with both modalities, we introduce **MuSP-Bench**, a human-authored benchmark of 490 questions targeting understanding across **Mu**sical **S**cores and **P**erformances. The benchmark distinguishes itself through long-horizon, open-ended questions spanning score, performance, and interpretive reasoning across classical piano and orchestral works. We evaluate frontier multimodal large language models under multiple input conditions and formats. Our results show that these models struggle substantially to understand scores, while facing even greater challenges when reasoning about performance audio. The benchmark is available at musp.vaclis.net.

1. INTRODUCTION

Musicians commonly express musical ideas through both scores and performances. Scores encode intent—pitch, rhythm, dynamics, articulation, and structure—while performance realizes that intent through time and tone. Musicians continually reason over these representations when interpreting, imagining, performing, and listening. True artificial musical intelligence capable of supporting real-world applications in education or artistic practice must therefore integrate score and performance understanding, reasoning both within and across these modalities and their common representations: audio, MIDI, ABC, and image.

Existing benchmarks largely assess these modalities in isolation. For instance, MusicTheoryBench [1] evaluates understanding of mainly short ABC excerpts; MMMU [2] and WildScore [3] focus primarily on single-image scores; ZIQI-Eval [4] mostly probes factual knowledge, though it includes an ABC continuation task; ABC-Eval [5] is limited to ABC; and all predominantly use multiple-choice evaluation. MSU-Bench [6] evaluates complete scores, but predominantly through short-horizon or binary questions. MusiXQA [7] and SSMR-Bench [8] generate synthetic scores and questions; while useful for symbolic understanding, they provide limited coverage of higher-level musical interpretation.

On the other hand, benchmarks like HumMusQa [9], MUSE [10], CMI-Bench [11], and MusTBench [12] assess musical perception from audio, including pitch, genre, melody, and instrumentation, as well as temporal grounding. However, none of these benchmarks assess understanding of performance in relation to the score or how interpretative choices shape the realization of a work. MuseBench [13] tests reasoning over score and audio, but its audio-input tasks mainly ask about key accuracy, completeness, tempo stability, and speed through true/false questions; as such, it lacks open-ended questions about voicing, technique, or expression.

Overall, existing benchmarks rarely evaluate long-horizon reasoning across score and audio or address the interpretive questions musicians ask: how themes unfold, where interpretive and technical challenges arise, and how a performance relates to the written score. We therefore introduce a multimodal benchmark that combines local musical understanding with long-horizon reasoning and real-world inquiries across score and performance.

2. BENCHMARK

MuSP-Bench comprises 490 questions across 18 classical piano pieces or movements and 6 orchestral excerpts. Two professional musicians authored questions requiring information from the score, performance, or both. Of the 490 questions, 460 questions are open-ended within a predefined answer format, such as a pitch, chord progression, timestamp, or phrase, and may allow multiple correct answers to reflect musical ambiguity; 30 provide multiple options when a sufficiently precise format could not be specified. Piano scores come from (n)ASAP [14], corrected using consensus from PianoCoRe performance MIDI [15]; orchestral scores and audio come from BSED [16].

Each question is annotated along four dimensions. **Modality** indicates whether the minimal required information comes from the *score* (S), *performance* (P), *both* jointly (S&P), or *either* (S/P). **Horizon** indicates the minimum evidence span: *short* (19.2%; 1–2 bars or $< \pm 5$ s), *medium* (25.1%; 3–16 bars or approximately 5–30 s), *long* (36.1%; larger or noncontiguous), or *any* (19.6%; any part can be used as evidence). Pieces average 10 score pages (1–40) and ± 5 minutes (0:17–12:29). **Content** follows a hierarchy from *pitch*, through *temporal organization*, *or-*

chestration, and *performance realization*, to *melodic*, *harmonic*, *formal*, *interpretive*, and *contextual* understanding. **Action** specifies the operation: identification, localization, quantification, comparison, ordering, inference, transcription, explanation, or summarization.

2.1 Question Overview

Questions cover score analysis (S), performance understanding (P), score–performance relationships (S&P), and contextual recognition (S/P), including harmony, form, phrasing, voicing, sonic layers, rubato, motifs, themes, articulation, pedaling, fingering, texture, style, tonality, composer, and work identification. Examples include:

- S&P: Consider the theme starting at 2:51. In which bar does the performer reach the subverted, silent climax of this theme, characterized by a significant slowdown in tempo and loudness?
- S/P: In the final gesture, which pitch is projected like a voice rising above the dark surrounding texture to make a forceful, attention-commanding proclamation?
- P: How many F3 bass notes are played truly staccato in the performance before 0:08 (i.e. not elongated in any way)?

3. EXPERIMENTS

We evaluate 5 frontier multimodal foundation models using the input modalities they support: GPT 5.6 Sol [17] and Qwen 3.6 Plus [18] on text and score images, Audio Flamingo Next [19] on audio, and Muse Spark 1.2 [20] and Qwen 3.5 Omni Plus [21] on all three formats. For S or S/P, we use ABC notation and PDF scores converted to images; for P and S/P, audio and MIDI rendered as text with ABC pitch notation; for S&P, score images with audio and ABC with audio or MIDI. We remove metadata from all formats. A language-only baseline measures how far models can answer from the composer and title alone; see *Metadata* in Table 1. An additional ablation tests how note representation within MIDI affects performance. Reported scores use deterministic answer normalization.

4. RESULTS

On score-only questions, GPT and Muse Spark perform best with ABC (51.1% and 31.7%). The substantial advantage of ABC suggests that these models process structured symbolic notation more effectively than images. Performance-only results similarly yield remarkable results with MIDI-as-text: GPT reaches 59.4% and Muse Spark 30.2%; Qwen 3.5 performs better on audio than MIDI. On S&P questions, Muse Spark falls from 30.6% with ABC+MIDI to 6.9% with either image+audio or ABC+audio, while Qwen 3.5 remains below 6% across all three formats. The separated S/P subsets show that the value of each modality depends on the task. Symbolic input is overall strongest for general analytical questions, with GPT reaching 50.0% using MIDI-as-text and 44.4% using ABC. Audio and images perform well on global attributes: Qwen 3.5 reaches 79.2% with audio on tonality/style, while Muse Spark reaches 91.7% with images. Images also help with composer/composition questions, where GPT and Muse Spark score 58.3% and 79.2%, respectively. The ablation shows that replacing ABC pitch names with MIDI note values reduces performance for all models except Muse Spark; results here. Overall, GPT is the strongest model, while visual and audio inputs offer particular benefits for stylistic and contextual recognition.

Table 1. Accuracy (%) by subset and input condition. The #Q column gives the number of eligible questions. Bold marks the highest result in each row; underlining marks each model's highest non-metadata result within that subset. N/A denotes an unsupported input format or task. The final aggregates weight the 6 subsets by their respective numbers of eligible questions (excluding *Metadata*).

Subset	Input	#Q	GPT5.6	Qwen3.5	MuseSpark	Qwen3.6	Flamingo
S	Metadata	180	**5.0**	2.2	4.4	1.7	N/A
	ABC	180	**51.1**	11.7	31.7	11.1	N/A
	Image	180	**27.2**	12.2	15.0	13.3	N/A
P	Metadata	106	**6.6**	4.7	**6.6**	5.7	3.8
	MIDI-as-text	106	**59.4**	11.3	30.2	10.4	N/A
	Audio	106	N/A	**14.2**	3.8	N/A	0.9
SP	Metadata	72	**6.9**	4.2	4.2	1.4	N/A
	ABC+MIDI	72	**45.8**	5.6	30.6	6.9	N/A
	Image+audio	72	N/A	5.6	**6.9**	N/A	N/A
	ABC+audio	72	N/A	4.2	**6.9**	N/A	N/A
General (S/P)	Metadata	36	**13.9**	5.6	8.3	5.6	2.8
	ABC	36	**44.4**	16.7	33.3	11.1	N/A
	MIDI-as-text	36	**50.0**	13.9	36.1	11.1	N/A
	Image	36	**19.4**	16.7	11.1	**19.4**	N/A
	Audio	36	N/A	**16.7**	11.1	N/A	0.0
Tonality/ style (S/P)	ABC	48	**77.1**	50.0	68.8	52.1	N/A
	MIDI-as-text	48	**77.1**	39.6	54.2	35.4	N/A
	Image	48	68.8	50.0	**91.7**	47.9	N/A
	Audio	48	N/A	**79.2**	20.8	N/A	22.9
Composer/ title (S/P)	ABC	48	35.4	14.6	**54.2**	8.3	N/A
	MIDI-as-text	48	**31.2**	0.0	27.1	0.0	N/A
	Image	48	58.3	16.7	**79.2**	20.8	N/A
	Audio	48	N/A	**54.2**	2.1	N/A	0.0
	Metadata	394	**6.6**	3.6	5.3	3.0	3.5 [/142]
	ABC	312	**51.9**	18.6	41.0	17.0	N/A
	MIDI-as-text	238	**55.9**	15.1	35.3	13.4	N/A
	Image	312	**37.5**	19.2	36.2	20.5	N/A
	Audio	238	N/A	**35.7**	8.0	N/A	5.0
	ABC+MIDI	72	**45.8**	5.6	30.6	6.9	N/A
	Image+audio	72	N/A	5.6	**6.9**	N/A	N/A
	ABC+audio	72	N/A	4.2	**6.9**	N/A	N/A
Aggregate	**Weighted mean**	**490**	**48.1**	16.5	25.9	14.1	N/A
Aggregate	**Weighted max**	**490**	**55.3**	22.7	42.1	16.7	N/A

5. CONCLUSION

We introduce MuSP-Bench, a human-authored benchmark for analytical and interpretive reasoning across musical scores and performances. MuSP-Bench provides a foundation for tracking progress toward musical understanding across modalities. Model performance is task-dependent: structured symbolic representations are strongest for detailed analytical reasoning, while images and audio can help with global stylistic and contextual recognition. Performance audio and image-based scores remain challenging overall, suggesting that success on broad attributes does not imply robust understanding of musical structure and expression.

6. REFERENCES

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